BAMBINOS.LIVE · HIRING ASSESSMENT · STAGE 2 — OPERATIONS

Paid is not Scheduled.

Re-engineering the Student Renewal → Class Scheduling pipeline: a leakage audit of the four tables, a zero-touch fast-track for clean renewals, a working reconciliation engine, and a 90-day rollout plan for 200+ teachers and a 10-person Ops team.

Candidate	**Inam Khan**
Registration No.	**12303473**
Role	**AI Program Manager (Operations)**
Prototype repo	**github.com/inam0980/Bambinos-here**
Data snapshot	**Official 5-table dataset · as of 04 Nov 2026, 09:00**

> The gateway already knows **who paid, how much, and for which child, class and teacher**. So stop waiting for a teacher's form and an Ops click to repeat it. Let the payment schedule the class, and send humans **only the cases that need judgement**.

11 of 12
paid renewals not yet scheduled, all waiting more than 12h

₹1.27 L
collected with no classes booked against it

67 h
oldest wait, on a clean same-batch renewal

8 of 12
cleared automatically by the prototype with no human touch

A · Leakage Audit
Failure modes, trace of all 13 invoices, broken SLAs

B · Re-engineered Flow
What to delete, the fast-track, exception routing

C · Prototype
LangGraph engine, reason codes, drafted messages

D · Run & Roll out
Morning board, ROI, SOP, 30-60-90 plan

EXECUTIVE SUMMARY

The pipeline is waiting on the wrong signal

Today a renewal moves only when a **teacher's form** arrives and an **Ops agent** checks it by hand. But the payment gateway already knows who paid, how much, and for which student, class and teacher. Steps 2 and 3 copy that information again, adding typos and delay. For a child continuing in the same batch, they add nothing.

11 / 12	₹1,26,850	27 min	16–48 h
paid, not scheduled (>12h)	revenue in limbo	median teacher form lag (teachers are *not* the bottleneck)	age of the 8 forms waiting in the Ops queue

Five things the data says

1. **The queue is where renewals die, not the teacher.** Teachers filed 10 of 12 forms within 50 minutes of payment. Once filed, 8 forms waited 16–48h, and the queue was not worked first-in-first-out: form 3303 (clean, filed Mon 09:30) is still pending, while 3305 and 3312, filed later, were approved.
2. **"Approved" is not the finish line.** Of 4 approved forms, only 1 child has classes booked. One booking failed as "*batch full*", but the child is one of the 6 students in that full batch. Another was approved for a weekend slot that does not exist.
3. **Re-typing creates the errors that Ops then checks for.** 25% of forms contain a duplicate, a wrong amount or a wrong invoice ID (84931 typed for 84913). One form was raised on a *failed* payment with only a UPI "processing" screenshot as proof.
4. **Contact-based matching breaks on real families.** A mother paying from her own phone and email, NRI numbers (+971, +1), and a parent name containing Cyrillic letters that look identical to Latin ones all break exact matching. The invoice's system IDs (student_id, user_id) never break.
5. **Nothing measures the parent's experience.** No stage timestamps, no owner per stuck case and no metric for *paid but class missed*. 16.7% of paid invoices have no form at all, so Ops doesn't know those children renewed.

What I propose

Let the payment drive
Gateway webhook → reconciliation engine. For same-batch renewals the teacher form and manual approval are **deleted**, not automated.

Humans handle only exceptions
18 reason codes, each with one owner, an SLA and a drafted message. Failed payments are never approved; partials and discounts go to Finance or the Ops Lead.

Never break a live class
A child keeps the current seat until the new term is booked. Shadow mode comes before switch-on, and a kill switch sends everything back to manual.

Prototype result on the official data: 8 of 12 open renewals auto-approved (67% straight-through), 192 class sessions generated ready to insert, 4 true exceptions routed to Finance, Ops Lead, Scheduler and Parent, and 22 WhatsApp/Slack messages drafted. **Projected impact at ~400 renewals a week:** ≈110 Ops hours a week saved (≈2.7 FTE), and same-batch payment-to-schedule time cut from **days to under 15 minutes**.

DELIVERABLE A · ROOT-CAUSE PROBLEM DEFINITION

Where students get stuck in the 4-table chain

The happy path needs three different people to act, in order, on the same day. Every handoff is a place where a paid child can be dropped. The numbers in each box are the official dataset.

Step 1 · Parent / Gateway
invoices
13
12 paid/partial · 1 failed

→

Step 2 · Teacher re-types
faculty_form_unapproved
12
2 invoices with no form · 1 duplicate

→

Step 3 · Ops verifies by hand
booking_requests
4
8 forms still pending

→

Step 4 · Ops / Parent picks slot
class_schedule_bookings
1
student actually scheduled

① **Orphaned invoice.** Paid, but the teacher never filed. Invisible to Ops (84906, 84910).

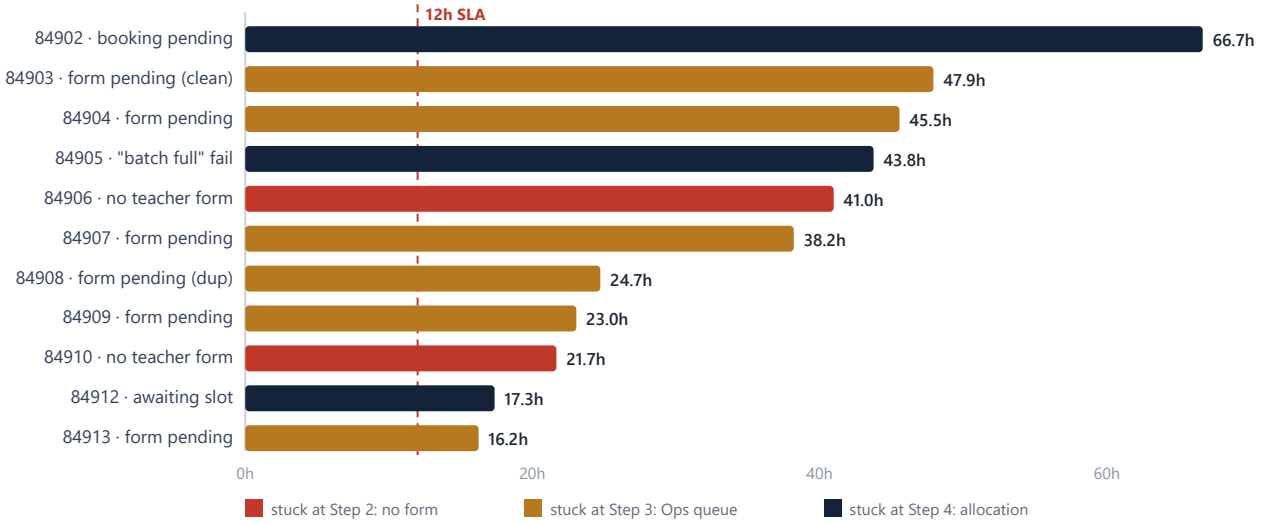
② **Dirty input.** Wrong invoice ID, wrong amount, duplicate, screenshot instead of payment.

③ **Queue stall.** No FIFO, no SLA, no weekend cover. Clean cases wait behind messy ones.

④ **Allocation bugs.** False "batch full"; batch change approved with no feasible slot.

⑤ **Class gap.** The old term ends before new bookings exist. The parent paid, the child misses class.

How long each paid renewal has been waiting (hours since payment, as of 04 Nov 09:00)



84901 is the only success: paid Sunday 10:15, scheduled 1.7h later because teacher and Ops happened to be online. 84911's payment failed, so it is not counted as paid.

Teacher → form

27 min median (15–50 min). Teachers are fast. The real Step-2 risk is the **17% who never file**.

Form → Ops approval

When it happens: 0.75–2.5h. But **8 forms sit 16–48h**, and the order of approval ignores arrival time.

Approval → scheduled

1 of 4 approved bookings became classes. Allocation is a second, unmonitored queue.

DELIVERABLE A · LEAKAGE AUDIT

All 13 invoices, traced through all four tables

Every row of the official dataset was joined across the four tables and the roster. Each one hides a different real-world failure mode.

Invoice	Where it sits	What actually went wrong	Root cause	Severity
84901	Scheduled · 1.7h	The only success. Sunday payment; teacher and Ops both online.	Success depends on luck and three people being available.	Baseline
84902	Booking pending · 66.7h	Approved Sunday 17:00. A same-batch child is still waiting for someone to "choose" the batch they are already in.	Step 4 is manual even when there is nothing to choose.	High
84903	Form pending · 47.9h	A perfectly clean case: IDs, amount and batch all match. Filed 25 min after payment, then untouched for 2 days while newer forms were approved.	100% manual review and no FIFO queue.	High
84904	Form pending	Mother paid from her own number (...XXXX77) and email (gmail vs outlook). Phone-based lookup fails. The teacher's note explains it, but nobody reads it.	Matching on contact details instead of system IDs.	Medium
84905	Booking **Failed_Batch_Full**	Batch 9205 shows 6/6, but one of those 6 is this child. A renewing student is refused their own seat.	Capacity rule doesn't exclude the renewing student.	High · bug
84906	No form · 41h	**Orphaned invoice.** GCC parent paid ₹18,000. No downstream row exists, so Ops can't see it.	The flow starts only when the teacher remembers.	High
84907	Form pending	Teacher gave a 20% "retention scholarship" (₹1,900 off). No approval trail and no policy limit.	Discount set at the link stage with no guardrail.	Medium · ₹
84908	2 forms pending	The same renewal was submitted twice, 5 minutes apart ("browser froze"). Risk of a double booking or double count.	No idempotency on the form.	Medium
84909	Form pending	Installment 1 of 2. The gateway captured ₹5,250, but the teacher typed ₹10,500.	Teacher re-types money data the gateway already has.	Medium · ₹
84910	No form · 21.7h	Second orphaned invoice (Chess, Coach Arvind).	Same as 84906.	High
84911	Form pending	Payment **Failed** (no payment_id), yet a form was filed with a "UPI processing" screenshot. A tired agent could approve unpaid classes.	The form lets a screenshot stand in for gateway proof.	High · ₹ risk
84912	Awaiting parent slot	Ops approved a move to a Sat-Sun 11:00 batch, but class 301 has no weekend batch. The parent is asked to choose from options that don't exist.	Approval happens before a feasibility check.	Medium
84913	Form pending	Teacher typed invoice **84931** (digits swapped). The parent name on the invoice contains a **Cyrillic "Me"** that looks identical to Latin "Me", so text search silently misses it.	Free-text keys and no Unicode normalisation.	Medium

Failure-mode taxonomy (this sample plus what to expect at scale)

Trigger
Teacher forgets, is on leave, or quits mid-term. Weekend and late-night payments wait for Monday (2 of 13 paid on a Sunday).

Identity
A different payer phone or email, siblings under one account, NRI formats, homoglyphs, masked PII, re-used numbers.

Money
Partial or installment payments, failed or pending UPI, unapproved discounts, FX (US/GCC), refunds and chargebacks after scheduling.

Data entry
Wrong invoice ID or link, wrong amount, wrong batch, duplicates, free-text remarks that carry critical facts.

Scheduling
A false "batch full", changes approved without a slot, parents who never answer the slot picker, teacher leave clashes.

Visibility
No stage timestamps, no owner, no SLA clock, no "paid but missed class" metric. The problem is felt by parents before Ops sees it.

DELIVERABLE A · QUANTIFY THE RISK

The metrics that are broken or invisible today

One North-Star metric that the parent feels, supported by leakage metrics that show *where* it breaks and *who* owns the fix. Every metric below is computed by the prototype from the four tables (`outputs/metrics.json`).

> **North Star: Payment → Class-Ready Time (P95).** Time from `invoices.paid_at` to the first `class_schedule_bookings.created_at`. Today: 1.7h for the single success; the other 11 are still open after 16–67h.
> **Target: same-batch ≤ 15 min, all renewals ≤ 24h.**

Metric	Definition (from the tables)	Today (sample)	Target / SLA	Owner
Paid-Not-Scheduled >12h	Paid or partial invoices with no schedule row after 12h; count and ₹	11 · ₹1,26,850	0 clean cases; exceptions within their SLA	Head of Ops
Orphaned Invoice Rate	Paid renewals with no faculty_form row	16.7%	Not applicable after redesign (form not required)	Engine
Straight-Through Rate	Renewals scheduled with zero human touch	0%	≥60% by Day 60, ≥70% by Day 90	AI PM
Ops Queue Age (P95)	now − form.created_at for pending forms	47.5h	Exception queue ≤ owner SLA	Ops Lead
Queue Order Violations	Forms approved while an older eligible form waits	2 of 4	0 (oldest-first queue)	Ops Lead
Approval → Booking Conversion	Approved booking_requests that reach class_schedule_bookings	25%	100% within 15 min	Scheduler
Form Data Error Rate	Forms with wrong invoice, amount or a duplicate	25%	Form retired for same-batch renewals	Teachers
False Rejection Rate	Bookings failed for a reason that isn't true (e.g. false "batch full")	1 of 4	0	Tech / Ops
Unpaid-Approval Risk	Forms raised on Failed or Partial payments	2	0 approved without gateway "Paid"	Engine
Class-Gap Incidents	Paid students with a missed class between terms	Not tracked	0	Head of Ops
Exception SLA Breach %	Blocking exceptions older than their owner's SLA	Not tracked	<5%	Each owner

SLA ladder by owner

Engine · 15 min
Clean same-batch renewal, from webhook to booked sessions to parent confirmation. Runs 24×7.

Ops Lead · 4h
Discount approvals, identity mismatches, false "batch full" bug triage.

Finance / Teacher · 12h
Partial payments, fee mismatches, invoice not found, batch mismatches.

Scheduler / Parent · 24h
Batch changes, new-batch openings, failed-payment retries. The child stays in the current batch meanwhile.

DELIVERABLE B · PROCESS RE-ENGINEERING

Delete before you automate

Automating today's four steps would just make the wrong process run faster. For each step, I asked why it exists and whether that reason is still true.

Step today	Why it exists	Is that still true?	Redesign
2 · Teacher Renewal Form	Tells Ops who paid and for which batch	No. The invoice already carries student_id, user_id, class_id, educator_id and the gateway amount. The form re-types them, and 25% of forms carry errors.	**DELETED** for same-batch. Becomes an optional one-tap "change slot / add note".
3 · Ops manual approval	Confirms the payment is real	No. A signed gateway webhook with payment_id and status Paid is stronger proof than a person opening a link or a screenshot.	**DELETED** for clean cases. Rules approve; humans see exceptions only.
4 · Batch selection	Picks days and time	Only for a batch change. A same-batch renewing child already holds a seat.	**AUTO** current batch. Parent self-serve, showing only feasible slots, for changes.
0 · Payment link	Teacher sends a generic link	This is where the data problems start: no batch tag, free discounts.	**STRENGTHENED** Link generated per student and batch, discount ≤10% guard, installment plan tagged.

Target workflow: who touches a clean renewal? Nobody.

Teacher	Taps "Send renewal link" (pre-tagged student · batch · fee) … ~~Fill renewal form~~ … Gets FYI: "renewed, nothing to do"
Parent	Pays (UPI / card / Stripe) → WhatsApp: confirmed + first class date … Only if a change is wanted: picks from feasible slots
Gateway	Webhook: Paid / Partial / Failed → invoices row
Recon Engine	Match on IDs → Validate payment · fee · identity · batch → Decide → Write booking + 24 sessions → Draft messages
Ops (exceptions)	~~Verify every form~~ ~~Pick batch for every booking~~ → Works only "My exceptions" queue: reason code, evidence, one-click action, SLA clock
Head of Ops	09:00 Morning Board: stuck >12h, exceptions by owner, straight-through %, hours saved

Dashed and struck-through steps are removed. Green steps are automated. Amber steps are the only human touch points left.

Fast-track eligibility: all five gates must pass

① Paid — gateway status Paid with a payment_id. Screenshots are ignored.

→

② Identity — student_id + user_id + class + educator match the roster

→

③ Fee — net = standard fee, or discount ≤10%

→

④ Batch — same batch (seat already held) or a feasible matched change

→

⑤ Unique — no duplicate or conflicting booking

Contact mismatches, a typo in the teacher's invoice reference, a duplicate form or a missing form do **not** block. They are corrected automatically and logged as advisory notes, because the system IDs already prove who paid.

DELIVERABLE B · SMART EXCEPTION HANDLING

Each exception goes to the one person who can fix it

"Send it to Ops" is not routing. Each reason code names one owner, one SLA, one channel and one recommended action. The message is drafted before a human opens the case.

Situation (dataset case)	Rule	Who is alerted, and how	Outcome
Parent paid from a different phone/email (84904)	System IDs match, so identity is proven	Parent on WhatsApp: "Reply YES to add this number". Ops FYI only.	Auto-approved
Teacher typed the wrong invoice ID (84913)	Unique fuzzy match on student + user + class + amount	Teacher FYI: "we fixed 84931 → 84913".	Auto-approved
Paid, but no teacher form (84906, 84910)	Paid + IDs match roster → same batch	Teacher FYI: "no form needed; reply if the parent wants a new slot".	Fast-tracked
Duplicate form (84908)	Same student + invoice + amount	Teacher FYI: duplicate closed.	Earliest kept
False "batch full" (84905)	Renewing child already holds a seat	Scheduler (Slack) + bug ticket to fix the capacity rule.	Re-allocated
Teacher's discount 20% (84907)	Discount >10% teacher limit	Ops Lead on Slack with APPROVE / REJECT. SLA 4h.	Exception
Installment 1 of 2, teacher typed full amount (84909)	Partially_Paid; gateway amount wins	Ops Finance on Slack (12h). Parent gets a receipt and an installment-2 note.	Exception
Batch change to a slot that doesn't exist (84912)	No class-301 batch on Sat-Sun 11:00	Scheduler (24h): open a batch or offer the nearest slot. Parent is reassured the child stays in the current batch.	Exception
Failed payment + screenshot (84911)	Gateway ≠ Paid → never approve	Parent gets a retry link. Teacher is told "screenshot ≠ payment, no action needed".	Blocked

Messages the engine drafted for these cases (real output from `outputs/3_followup_messages.md`)

WHATSAPP → PARENT · PAYER_CONTACT_MISMATCH

Hi Ri***u Iy***r, we received INR 12,000 for An***a Iy***r's renewal, and classes are confirmed. The payment came from +91-9844XXXX77, which isn't on the profile. Reply YES to add it as a secondary contact for class reminders.

WHATSAPP → PARENT · PAYMENT_FAILED

Hi An***i Gu***a, we tried to confirm Is***n Gu***a's renewal but the payment of INR 12,000 did not go through. Please retry here: bambinos.live/invoice/BMB-INV-2026-84911. Is***n Gu***a's current classes continue meanwhile.

WHATSAPP → TEACHER · INVOICE_REF_AUTOCORRECTED

Hi Ms. Mary M, small fix: form 3313 had invoice 84931; the correct one is 84913. We've corrected it and renewed De***a Me***a. Nothing else needed.

SLACK → OPS LEAD · DISCOUNT_NEEDS_APPROVAL

```
:moneybag: RNW-84907 | Ms. Simran D gave Vi***n Jo***i a discount: paid INR 7,600 vs list INR 9,500. Above 10% teacher limit. Reply APPROVE / REJECT. SLA 4h.
```

SLACK → OPS FINANCE · PARTIAL_PAYMENT

```
:warning: RNW-84909 | Ad***t Ku***i paid installment 1 (INR 5,250) of INR 10,500. Teacher form says INR 10,500. Decide: approve under installment policy or hold. SLA 12h.
```

SLACK → OPS SCHEDULER · FALSE_BATCH_FULL

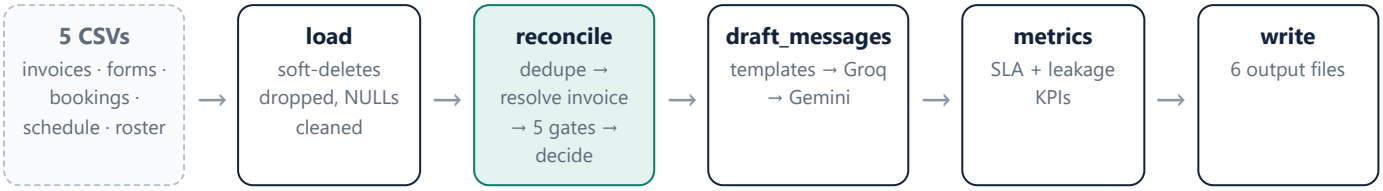
```
:white_check_mark: RNW-84905 | Booking for Re***h Pa***l failed as 'batch full', but Re***h Pa***l already holds a seat in CM-G4-Batch-D. Auto-reallocated. Please fix the capacity rule.
```

AI guardrail. The LLM never decides. Approve/reject is deterministic Python (`src/rules.py`): auditable, testable, the same answer every time. Groq or Gemini only rewrites the tone. Code checks that every amount, ID and date in the template is still in the rewrite, and if any is missing the template is sent instead. Provider chain: **Groq → Gemini → template**. A failing provider (bad key, quota, removed model) is skipped automatically.

DELIVERABLE C · WORKING PROTOTYPE (SUMMARY, DETAILS IN THE REPO)

Reconciliation & Auto-Approval Engine

Python + Pandas rules engine orchestrated as a **LangGraph StateGraph**, with LLM message polishing and a Streamlit Ops board. It ingests the five official CSVs unchanged. Run: `python src/pipeline.py` · `python -m streamlit run dashboard/app.py`



Results on the official dataset

Invoice	Decision	Reason codes
84901	Already scheduled	none
84902	Auto-approve	STUCK_AT_ALLOCATION
84903	Auto-approve	CLEAN
84904	Auto-approve	PAYER_CONTACT_MISMATCH
84905	Auto-approve	FALSE_BATCH_FULL
84906	Auto-approve	ORPHANED_INVOICE
84907	Exception	DISCOUNT_NEEDS_APPROVAL
84908	Auto-approve	DUPLICATE_FORM
84909	Exception	PARTIAL_PAYMENT, TEACHER_AMOUNT_MISMATCH
84910	Auto-approve	ORPHANED_INVOICE
84911	Exception	PAYMENT_FAILED
84912	Exception	BATCH_CHANGE_NO_MATCH
84913	Auto-approve	INVOICE_REF_AUTOCORRECTED, NAME_ENCODING_ANOMALY

Outputs (required by the brief)

Required output	File · count
(1) Auto-approved entries for class_schedule_bookings	`1_auto_approved…csv` · 8 `1b_class_schedule…insert.csv` · 192 sessions
(2) Flagged exceptions with exact reason codes	`2_flagged_exceptions.csv` · 13 lines (4 blocking cases) with owner, SLA and action
(3) Drafted WhatsApp/Slack follow-ups	`3_followup_messages.csv / .md` · 22
Full ledger + KPIs	`0_reconciliation_ledger.csv`, `metrics.json`

Design choices worth defending

- **Gateway is truth.** Form amounts and screenshots never override it.
- **Match on IDs, not contacts.** Phone and email differences are advisory only.
- **Fuzzy fallback** only when the match is *unique*; otherwise it's an exception.
- **Homoglyph normalisation** before any text comparison.
- **Seat-aware capacity:** a renewing child never counts against their own seat.
- **Idempotent:** re-running produces the same decisions, and duplicates are closed, not re-booked.
- **Works offline:** no API key is needed to run or review it.

DELIVERABLE D · LIVE OPS DASHBOARD

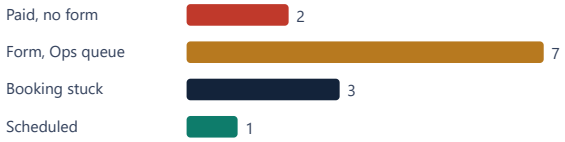
The Head of Ops' 9 AM board

One screen that answers three questions in 30 seconds: *Is any paid child at risk of missing class? Who owns each stuck case? Is automation actually saving time?* A working version is in `dashboard/app.py`; values below are from the official dataset.

Renewals → Scheduling · Morning Board — Wed 04 Nov 2026 · 09:00 · last sync 2 min ago

Paid, not scheduled	Stuck >12h	Auto-approved today	Open exceptions	Straight-through	Ops hours saved
11	11	8	4	67%	2.7h
₹1,26,850 in limbo	oldest 66.7h · 84902	192 sessions booked	0 past SLA	target 70%	today · 8 × 20 min

Where every paid renewal sits



Before the engine runs. After: 9 scheduled, 4 in exception queues.

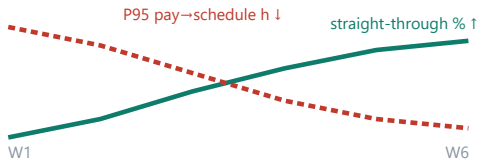
Blocking exceptions by owner · SLA clock

Ops Lead	84907 discount 20%	4h
Ops Finance	84909 installment 1/2	12h
Scheduler	84912 no Sat-Sun slot	24h
Parent	84911 payment failed	24h

Red list: oldest first, one-click actions

Case	Reason	Age	Action
RNW-84907	DISCOUNT_NEEDS_APPROVAL	38.2h	[Approve] [Reject]
RNW-84909	PARTIAL_PAYMENT	23.0h	[Installment OK] [Hold]
RNW-84912	BATCH_CHANGE_NO_MATCH	17.3h	[Offer slots] [Open batch]

Weekly trend (illustrative targets)



Teacher league (coaching, not blame)

Per teacher, last 7 days: renewal links sent · paid · conversion % · data issues · avg discount. Example today: Acharya Nancy S. sent 3, 1 duplicate form, 1 infeasible batch request. Ms. Simran D. had 1 discount above the limit and 1 form on a failed payment.

Refresh
Event-driven on gateway webhook, plus a 15-minute reconciliation sweep. "Last sync" is shown so a dead pipe is visible.

Alerts
Slack @owner on new blocking exceptions. Escalation to the Ops Lead at 75% of SLA, and to the Head of Ops at breach.

Stack
Prototype: Streamlit. Production: Metabase on the MySQL tables, with the same metric definitions as `metrics.json`.

DELIVERABLE D · DATA, TIME & COST SAVINGS

What this is worth: a realistic estimate

The dataset is a sample, so volumes are assumptions. I show the model so the Head of Ops can replace each input with real numbers in Week 1.

Effort per renewal today vs. after

Activity	Today	After
Teacher fills renewal form	6 min	0 (same-batch)
Ops verifies invoice vs form	8 min	0 (engine)
Ops allocates batch, creates bookings	6 min	0 (engine)
Chasing on exceptions (~35% × 15 min)	5.3 min	—
Exceptions, pre-diagnosed (~33% × 8 min)	—	2.6 min
Ops minutes per renewal	**19.3**	**2.6**

Assumptions to validate with a time-and-motion sample in Days 0–30. The 33% exception rate is taken from this dataset (4 of 12), which was built to be messy, so production is likely to be lower.

Sensitivity: Ops hours saved per week

Renewals / week	Ops h saved	≈ FTE (40h)	Teacher h saved
200	55 h	1.4	14 h
400 (base)	**110 h**	**2.7**	**28 h**
800 (peak / term start)	221 h	5.5	56 h

Teacher hours assume 70% of renewals are same-batch × 6 min of form-filling.

> **The bigger value isn't hours.** It's the ₹1.27L of paid revenue sitting unscheduled in a 3-day sample, and the parents who would ask for refunds after a missed class. Cutting same-batch time from days to minutes protects renewal NPS and next-term conversion.

Before → after, at a glance

Measure	Today (sample)	Day 90 target
Payment → class-ready, same batch	1.7h best case; 16–67h for the rest	≤15 min P95, 24×7, including weekends
Human touches per clean renewal	3 (teacher, Ops verify, Ops allocate)	0
Straight-through rate	0%	≥70%
Paid-not-scheduled >12h	11 (₹1.27L)	0 clean cases; exceptions within SLA
Approvals on unpaid/partial without review	Possible (84911 screenshot)	Impossible by rule
Run cost	~3 FTE of manual checking	Hosting + LLM tokens for tone only (optional; templates are free)

DELIVERABLE D · SOP & CHANGE MANAGEMENT

Rolling out without breaking a single live class

> **Prime directive:** no existing student is ever removed from their current batch until the new term's sessions are written and confirmed. Every mode (manual, shadow, auto) obeys this. Class gaps are prevented by design, not by vigilance.

SOP: who does what in the new world

Role	Daily routine	Never do
Ops Agent (×6)	Work only the "My exceptions" queue, oldest first. Each card shows the reason code, evidence and a drafted message. Resolve with one click (approve / send / reassign). Log a reason on every override.	Approve anything the gateway doesn't show as Paid. Accept screenshots.
Ops Scheduler (×2)	Batch changes, new-batch openings, FALSE_BATCH_FULL tickets, teacher-leave clashes. Offer the parent only feasible slots.	Move a child before the new seat is booked.
Ops Lead / Finance (×2)	Discount approvals (4h), installment decisions (12h), a weekly 30-min rule review from the override log.	Change rules without a shadow-test diff.
Teacher (200+)	Send the renewal link from the app. No form. Reply to the WhatsApp nudge only if the parent wants a different slot.	Edit amounts, or create custom discounts above 10%.
Weekend on-call (×1, rotating)	The engine handles same-batch 24×7. On-call covers blocking exceptions with a 12h weekend SLA.	—

Change management for 200+ teachers

- **The pitch is "less work":** "You no longer fill the renewal form." Removing a task needs little persuasion.
- **Pilot with 20** high-renewal teachers from different subjects and regions (India, GCC, US), including the ones with the most form errors.
- **Train in 10 minutes:** a WhatsApp voice note, a 60-second screen recording and a pinned FAQ. No classroom training.
- **Feedback loop:** teachers see their renewals confirmed in real time. The Teacher League recognises good performers and doesn't publicly name poor ones.

Change management for the 10-person Ops team

- **Role shift, not role cut:** from "verifier" to "exception specialist". No redundancies in the first 90 days; freed time goes to parent experience and batch planning.
- **Two Ops champions** co-own the rule book and sign off every new reason code.
- **Shadow mode builds trust:** agents see the engine agree with them ≥98% of the time before it acts alone.
- **Kill switch:** one flag sends everything back to the manual queue. Auto-approvals are reversible for 24h.

DELIVERABLE D · 30-60-90 DAY PLAN

See → Fast-track → Remove

Each phase has an exit gate. We move on only when the numbers say it's safe.

DAYS 0–30

See & shadow

- Gateway webhook → engine. Stage timestamps on all 4 tables.
- Morning Board live for the Head of Ops.
- **Quick wins:** fix the false "batch full" capacity bug; enforce an oldest-first Ops queue.
- Engine output used to **clear today's backlog** (the 11 stuck renewals).
- Shadow mode on 100% of renewals. Time-and-motion baseline.

Exit gate: ≥98% engine–human agreement over 2 weeks · zero paid-not-scheduled >72h · board used daily.

DAYS 31–60

Fast-track on

- Same-batch auto-approve **switched on** for 20 pilot teachers, then 100.
- Slack one-click actions. Meta-approved WhatsApp templates.
- LLM tone layer (Groq → Gemini → template) with number guardrails.
- Form made optional for same-batch renewals.
- Weekly rule review from the override log.

Exit gate: straight-through ≥60% · P95 same-batch ≤1h · class-gap incidents = 0 · no wrong auto-approvals.

DAYS 61–90

Remove & scale

- Per-student renewal links with a discount guard and installment tags.
- Parent self-serve slot picker showing **only feasible batches**.
- All 200+ teachers. The **teacher form is retired**.
- Ops reorganised into exception pods (Finance · Scheduling · Identity).
- Metabase production dashboard. SLA alerts.

Exit gate: straight-through ≥70% · P95 ≤15 min · Ops hours on renewals −60% · renewal CSAT ≥4.5.

Risk register

Risk	Mitigation	Residual
Wrong auto-approval	Approve only on gateway Paid + exact ID match. Shadow proof first. Full audit log. 24h reversal window.	Low
Webhook outage / missed event	15-minute polling sweep. "Last sync" on the board, with an alert if older than 30 min.	Low
LLM writes a wrong fact	LLM only rewrites tone. Numbers are verified in code, templates are the fallback, and no decisions are made by the LLM.	Low
Teachers keep using the old form	Form still accepted and reconciled as metadata. Engine ignores its trigger role. Gentle nudge.	Medium
Ops trust / job anxiety	Champions, shadow agreement data, an explicit no-redundancy commitment, new exception-specialist roles.	Medium
PII exposure	Masked data. Only IDs and first names go to the LLM. WhatsApp sends use approved templates only.	Low

Questions I'd bring to the Head of Ops in the live review

1. What share of renewals are same-batch today? This sets the real straight-through ceiling.
2. Does the payment link already carry classbatch_id, or only class_id? This decides how much Step 2 can be dropped on Day 1.
3. What is the policy on installments: schedule all 24 sessions, or only the paid half?
4. Who is allowed to give discounts today, and up to what %?
5. When does a renewal term start: the day after the last session of the current term, or a fixed calendar date?